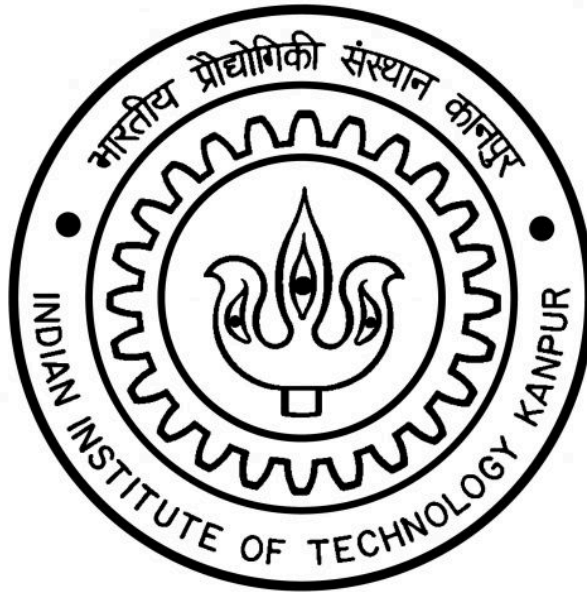




CE432

TERM PAPER REPORT

Demonstration of Geographic Information System (GIS) Functions Using Microsoft Excel

Instructors: Dr. Salil Goel , Maj Gen.(Dr.) Nagarajan Balasubramanian

Group 7

Submitted By:

Ansh Kumar (230158)

Anurag Sharma (230174)

Table of Contents

1. Introduction	3
2. Objectives	4
3. Dataset Description	4
4. Methodology & Results	5
4.1 Distance Calculation (Haversine Formula)	5
4.2 Select Within Distance	6
4.3 Select By Location	7
4.4 Nearest Neighbor Analysis	9
4.5 Slope Calculation	9
4.6 Bearing / Azimuth Calculation	11
4.7 Conditional Formatting	13
4.8 Pictorial Representation	15
4.9 Choropleth Map using Excel 3D Maps (Bing Maps)	17
4.10 Coordinate Mapping using ArcGIS for Excel(Add-in)	18
4.11 Heat Map Visualization	19
5. Discussion & Critical Analysis	20
6. Conclusion	21
7. References	22

1. Introduction

A geographic information system (GIS) is a computer system for storing, managing, analyzing, and displaying geospatial data. Since the 1970s GIS has been important for professionals in natural resource management, land use planning, natural hazards, transportation, health care, public services, market area analysis, and urban planning. GIS is a framework that integrates hardware, software, and spatial data for the purpose of creating, capturing, managing, analyzing, and displaying geographically referenced information. More recent integration of GIS with the Internet, global positioning system (GPS), wireless technology, and Web service has found applications in location-based services, Web mapping, in-vehicle navigation systems, collaborative Web mapping, and volunteered geographic information. GIS is the core of geospatial technology, which is related to a number of fields including remote sensing, Global Positioning System (GPS), cartography, surveying, geostatistics, Web mapping, programming, database management, and graphics design. The origins of GIS in its present form lie in the application of rapidly developing computing tools, especially computer graphics in a variety of fields such as urban planning, land management, and geocoding in the 1960s and 1970s. The first operational GIS is reported to have been developed by Roger Tomlinson in the early 1960s for storing, manipulating, and analyzing data collected for the Canada Land Inventory (Tomlinson 1984). In 1964, Howard Fisher founded the Harvard Laboratory for Computer Graphics, where several well-known computer programs of the past such as SYMAP, SYMVU, GRID, and ODESSEY were developed and distributed throughout the 1970s (Chrisman 1988). [5]

For civil engineering purposes, GIS applications range from site suitability inspection and transportation network analysis to flood inundation mapping. But the operational complexity of professional GIS platforms creates an obstacle for students and other users who don't have access to licensed software. Microsoft Excel, with its broad availability and formula transparency offers a feasible alternative for understanding and demonstrating the mathematical foundations of GIS operations.

This study inspects eleven GIS functions applicable within the Excel environment. While Excel cannot replicate the full visualization capabilities of dedicated GIS software, it can accurately compute some mathematical computations, spatial metrics and serve as a pedagogical tool for understanding the mathematics underlying GIS analysis. The functions are demonstrated using different datasets ensuring relevance and verifiability.

1.1 Scope of the Study

- Implementation of 11 GIS functions within MS Excel using built-in mathematical functions
- Use of Indian city coordinates, health facilities, traffic accidents datasets
- Mathematical derivation and Excel formula construction for each function
- Comparison of results with known reference values (Google Maps distances, etc.)
- Critical evaluation of Excel's capabilities and limitations for GIS analysis.

2. Objectives

The primary objectives of this term paper are:

1. To identify and document eleven fundamental GIS functions that can be mathematically implemented within Microsoft Excel.
2. To create a functional Excel workbook demonstrating all eleven GIS operations using real geographic datasets.
3. To critically analyze the potential and limitations of Excel as a GIS demonstration platform.
4. To document findings in the form of a comprehensive academic term paper suitable for CE432 submission.

3. Dataset Description

The primary dataset used in this study comprises geographic centroid coordinates of major cities of India. Coordinates were sourced from kaggle.com. A secondary dataset comprising health facilities in Ghana and a tertiary traffic accident dataset of Indian States for visualizing heat maps.

#	State	District	Latitude	Longitude
1	Andaman and Nicobar	Andaman Islands	12.3825	92.8229

Table 3.1 : Indian Districts Centroid Coordinates (Source: [Kaggle.com](https://www.kaggle.com))

Region	District	Facility Name	Type	Town	Ownership	Latitude	Longitude
Ashanti	Adansi South	Hwidiem Community Clinic	Clinic	Hwidiem	Government	5.89432	-1.46766

Table 3.2 : (Source: [Kaggle.com](https://www.kaggle.com))

district	std-c ode	state	gst-st ate-c ode	iso-366-2	population	rank	Latitude	Longitude	altitude
South Andam an	3192	Andaman and Nicobar Islands	AN	IN-AN	108058	423	11.62338	92.72648	16.12904

Sl. No.	State	Road Accidents - Cases	Road Accidents - Injured	Road Accidents - Died	Railway Accidents - Cases	Railway Accidents - Injured	Railway Accidents - Died	Railway Crossing Accidents - Cases	Railway Crossing Accidents - Injured	Railway Crossing Accidents - Died	Total Traffic Accidents - Cases	Total Traffic Accidents - Injured	Total Traffic Accidents - Died
1	Andhra Pradesh	21070	21340	8293	1029	0	1037	0	0	0	22099	21340	9330

4. Methodology & Results

4.1 Distance Calculation (Haversine Formula)

Definition

The Haversine formula computes the great-circle distance between two points on a sphere using their latitudes and longitudes. It is the standard formula used in GIS for straight-line geographic distance.[8]

Geographic distance computation is the most fundamental operation in GIS. While Euclidean (straight-line) distance is suitable for small areas on flat maps, it introduces significant errors over large distances due to Earth's curvature. The Haversine formula corrects for this by treating Earth as a sphere of radius $R = 6,371$ km.

Mathematical Derivation

Let Point 1 have coordinates (ϕ_1, λ_1) and Point 2 have coordinates (ϕ_2, λ_2) in radians.

$$d = 2R \arcsin \left(\sqrt{\sin^2 \left(\frac{\phi_2 - \phi_1}{2} \right) + \cos(\phi_1) \cos(\phi_2) \sin^2 \left(\frac{\lambda_2 - \lambda_1}{2} \right)} \right)$$

Where $R = 6,371$ km ϕ represents latitude in radians and λ represents longitude in radians. The formula accounts for spherical geometry and produces the shortest surface distance between two points.

Excel Implementation

1. Using xlookup formula in excel to get data of latitude and Longitude of target districts from other sheets where our data is.

Excel Formula

```
= IFERROR(XLOOKUP(city1, 'district wise centroids'!District_Names, 'district wise centroids'!Lat_coords), "Not Found")
```

2. Convert latitudes and longitudes from decimal degrees to radians using =RADIANS()
- 3.. Apply Haversine formula to get distance in km

Excel Formula

```
=2*6371*ASIN(SQRT(SIN((RADIANS(Lat2-Lat1))/2)^2 +  
COS(RADIANS(Lat1))*COS(RADIANS(Lat2))*SIN((RADIANS(Lon2-Lon1))/2)^2))
```

Sample Results

Point 1 City Name	Varanasi	Point 2 City Name	Bhilwara
Latitude (°N) of point 1	25.360570888888894	Latitude (°N) of Point 2	25.451197677595637
Longitude (°E) of Point 1	82.82529666666667	Longitude (°E) of Point 2	74.63947595628417
Distance (in Km)		822.1280153	

4.2 Select within Distance

Definition

Select within distance finds and selects features (points, lines, or polygons) that lie within a specified distance of a given reference feature or layer, using geographic distance as the proximity criterion.

This algorithm creates a selection in a vector layer. The criteria for selecting features is based on the spatial relationship between each feature and the features in an additional layer.

Select within distance is a vector spatial query operation in GIS used to identify features based on their geographic distance to other features. sometimes visual inspection may suggest that which features are nearby but GIS performs this analysis using pre-defined distance thresholds. The operation determines spatial relationships by measuring distances either Euclidean for the planar data or geodesic for the curved Earth surfaces and selects all features that lie within a specified radius of a reference layer. This enables accurate identification of nearby entities for analysis, planning, and decision-making.

Excel Implementation

1. Using xlookup for finding the latitude and longitude of the target district

Excel Formula

```
=IFERROR(XLOOKUP(target_district,District_col,Long_col), "Not Found")
```

2. Making a new column of measuring distance as we have done in the last function, here we doing for all columns with respect to target district
3. Using countif formula function to get counts of all the district in proximity distance

Excel Formula

```
=COUNTIFS(Distance_from_target, "<=" & proximity_distance, Distance_from_target, ">0")
```

4. Apply Conditional Formatting rule to highlight the selected districts.

Excel Formula

=AND(Distance_from_target <= proximity_distance, Distance_from_target>0)

5. Apply Filter formula in new column of selected districts

Excel Formula

=FILTER(District_Names, (Distance_from_target <= proximity_distance) * (Distance_from_target>0), "None within range")

Sample Results

Selection By Proximity	
Name of Target District	Rampur
Latitude of Target District	28.83244727272725
Longitude of Target District	79.10977333333337
Proximity Distance	1229
Total Selected Districts	418

State	District	Latitude	Longitude	Distance from Target	selected districts
Andaman and Nicob	Andaman Islands	12.382571069235052	92.82291050641506	2315.51673	Adilabad
Andaman and Nicob	Nicobar Islands	7.835290749774195	93.51160123757859	2779.597686	Karimnagar
Andhra Pradesh	Adilabad	19.284513950073435	78.81321233480176	1062.10709	Medak
Andhra Pradesh	Anantapur	14.312065561843147	77.46015845	1623.50791	Nizamabad
Andhra Pradesh	Chittoor	13.331092630676961	78.92763898886037	1723.774658	Warangal
Andhra Pradesh	Cuddapah	14.490289134125629	78.69613870967754	1595.343581	Barpeta
Andhra Pradesh	East Godavari	16.782718320784902	82.24320747	1377.613024	Bongaigaon
Andhra Pradesh	Guntur	15.884925767634849	80.58657614107885	1447.635196	Dhuburi
Andhra Pradesh	Hyderabad	17.38971555555556	78.46668888888888	1274.063634	Goalpara
Andhra Pradesh	Karimnagar	18.515483193277312	79.43765428571413	1147.677825	Kokrajhar
Andhra Pradesh	Khammam	17.683921325051784	80.67008643892335	1249.815548	Araria
Andhra Pradesh	Krishna	16.143872753151673	81.14805061000413	1426.265711	Aurangabad
Andhra Pradesh	Kurnool	15.43759715142429	77.90086836581717	1494.599653	Banka
Andhra Pradesh	Mahabubnagar	16.639546224256293	77.98933089244859	1360.621301	Begusarai
Andhra Pradesh	Medak	17.889304978038055	78.13778901903379	1220.841253	Bhabua
Andhra Pradesh	Nalgonda	17.113250515463907	79.16984453608244	1303.129657	Bhagalpur
Andhra Pradesh	Nellore	13.894344071421889	80.14702671812498	1664.481882	Bhojpur
Andhra Pradesh	Nizamabad	18.429485849056604	78.07523490566035	1161.526764	Buxar
Andhra Pradesh	Prakasam	15.489039001601705	80.11665790176184	1487.313547	Darbhanga
Andhra Pradesh	Rangareddi	17.245235872576178	78.05756828254849	1292.906282	Gaya
Andhra Pradesh	Srikakulam	18.570900998697432	84.25874814010675	1255.227976	Gopalganj
Andhra Pradesh	Vishakhapatnam	17.64683903	83.01014035778171	1305.761508	Jamui
Andhra Pradesh	Vizianagaram	18.283296960329718	83.45982854198873	1253.660196	Jehanabad
Andhra Pradesh	Warangal	18.01752608695651	79.87422028985509	1205.079364	
Andhra Pradesh	West Godavari	16.63297943925236	81.45912990654196	1377.628581	
Arunachal Pradesh	Changlang	27.360484793388455	96.37552644628106	1699.999416	

Fig 1 :Excel sheet showing Selected Districts within the proximity of target district

4.3 Select by location

Definition

Selection by location is a spatial query operation in GIS that selects features from a target layer that lie within a reference layer

This algorithm creates a selection in a vector layer. The criteria for selecting features is based on the spatial relationship between each feature and the features in an additional layer. Select by location is a vector spatial query operation in GIS used to identify features that fall inside a reference feature or layer. Instead of relying on visual estimation, GIS computes this relationship mathematically by measuring distances between geometries. It depends on the data model, for planar coordinate systems the Euclidean distance is used, while geodesic distance is preferred for large-scale or global analyses to account for Earth's curvature. The function effectively calculates whether the target feature lies inside or outside the reference feature or layer and selects all target features that lie within the reference layer, enabling precise analysis for planning and spatial decision-making.

Excel Implementation

1. Using Conditional Formatting to set rules for selection of districts within maximum minimum latitude and longitude

Excel Formula

```
=AND(Latitudes>= min_latitude, Latitude <= max_latitude, Longitude >= min_longitude, Longitude <= max_longitude)
```

2. Applying a filter formula to get the name of districts whose centroids lies in the box.

Excel Formula

```
=FILTER(Districts, (Latitude>=min_latitude) * (Latitude <=max_latitude) * (Longitude>=min_longitude) * (Longitude<=max_longitude), "None in area")
```

Sample Results

Select by location				
Min Latitude	13			
Max Latitude	20			
Min Longitude	70			
Max Longitude	95			
Total districts lies in this box	76			

State	District	Latitude	Longitude	Name of districts whose centriod lies in the box
Andaman and Nicobar	Andaman Islands	12.38257107	92.82291051	Adilabad
Andaman and Nicobar	Nicobar Islands	7.83529075	93.51160124	Anantapur
Andhra Pradesh	Adilabad	19.28451395	78.81321233	Chittoor
Andhra Pradesh	Anantapur	14.31206556	77.46	Cuddapah
Andhra Pradesh	Chittoor	13.33109263	78.92763899	East Godavari
Andhra Pradesh	Cuddapah	14.49028913	78.69613871	Guntur
Andhra Pradesh	East Godavari	16.78271832	82.24	Hyderabad
Andhra Pradesh	Guntur	15.88492577	80.58657614	Karimnagar
Andhra Pradesh	Hyderabad	17.38971556	78.46668889	Khammam
Andhra Pradesh	Karimnagar	18.51548319	79.43765429	Krishna
Andhra Pradesh	Khammam	17.68392133	80.67008644	Kurnool
Andhra Pradesh	Krishna	16.14387275	81.14805061	Mahbubnagar
Andhra Pradesh	Kurnool	15.43759715	77.90086837	Medak

Fig 2: Excel sheet showing Selected districts rows highlighted and filtered in new column

4.4 Nearest Neighbor Analysis

Definition

Nearest Neighbor analysis finds the closest feature (point, facility, or city) to a given query point from a set of known locations, using geographic distance as the proximity metric.

Nearest neighbor analysis is used in location-based services, network analysis, and facility planning. Given a set of cities and a query location, this function identifies which city is closest using the Haversine distance formula applied across all candidates, then identifies the minimum.

Mathematical Formula

For each point P_i : $d_i = \text{Haversine}(\text{Query}, P_i)$

$\text{Nearest} = \text{argmin}(d_1, d_2, \dots, d_n)$

Excel Implementation

1. Compute Haversine distance from the query point to every city and store it in a new column
2. Use `xlookup` to get latitude and longitude of target district
3. Use `Index` to get minimum within the column

Excel Formula

`=INDEX(City_Names, MATCH(MINIFS(Distance_Column, Distance_Column, ">0"), Distance_Column, 0))`

Sample Results

Nearest Neighbour	
City	Bhilwara
Latitude	25.451198
Longitude	74.639476
Nearest Neighbour	Rajsamand

4.5 Slope Calculation from Elevation Data

Definition

Slope is the rate of change of elevation with respect to horizontal distance. It is a fundamental terrain parameter computed in GIS from Digital Elevation Models (DEMs) and is critical for drainage design, erosion analysis, and road alignment.

Slope analysis is essential in civil and environmental engineering. Steep slopes create challenges for road construction, agricultural irrigation, and building foundations. In GIS, slope is computed from a DEM (Digital Elevation Model) by comparing elevation values at adjacent grid cells. In Excel, slope can be computed between any two points where elevation data is available.

Mathematical Formula

$$\text{Slope (\%)} = \left(\frac{\Delta \text{Elevation}}{\text{Horizontal Distance}} \right) \times 100$$

$$\text{Slope (degrees)} = \tan^{-1} \left(\frac{\Delta \text{Elevation}}{\text{Horizontal Distance}} \right)$$

Excel Implementation

1. Use xlookup to now get latitude, longitude and altitude of target district
2. Make a new column of distance of each city from target district using haversine formula as discussed before
- 3.. Use the formula for slope in percentage

Excel Formula

= (Z2-Z1)/(Dist_km*1000)*100

3. Use this to convert slope to degrees

Excel Formula

=DEGREES(ATAN((Z2-Z1)/(Dist_km*1000)))

Sample Results

Slope w.r.t City	
city	Rampur
Latitude	28.789304

Longitude	79.024952
Altitude	190.06732

city	latitude	longitude	altitude	Distance	Slope(%)	Slope(degree)
Port Blair	11.62338	92.72648	16.12904	2380.097502	0.007308	3.76752E-05
Adoni	15.63222	77.27284	425.7995	1474.004459	-0.0159926	-6.08933E-05
Amaravati	16.57452	80.35901	34.70644	1365.061311	0.0113812	6.56609E-05
Anantapur	14.68189	77.60059	348.9019	1575.505635	-0.0100815	-5.68956E-05
Bhimavaram	16.54471	81.52059	8.032158	1385.269376	0.0131408	6.47421E-05
Chilakaluripet	16.09243	80.16239	34.29998	1416.629494	0.0109956	6.32714E-05
Chittoor	13.21718	79.10033	310.4805	1731.559164	-0.006954	-5.17015E-05
Dharmavaram	14.42273	77.72147	366.4323	1603.119671	-0.0110014	-5.59379E-05
Eluru	16.71066	81.09524	21.15529	1359.664612	0.0124231	6.59433E-05
Gudivada	16.44103	80.99263	12.98032	1387.754245	0.0127607	6.46198E-05
Guntakal	15.16741	77.37362	456.1523	1524.159437	-0.0174578	-5.89077E-05
Guntur	16.30665	80.43654	39.71604	1395.511633	0.0107739	6.42194E-05
Hindupur	13.82907	77.49184	628.6644	1671.003899	-0.0262475	-5.37817E-05
Kadapa	14.46735	78.82413	137.4309	1592.662418	0.0033049	5.58258E-05

Fig 3: Excel sheet showing Result Table with slope(%) and slope(°)

4.6 Bearing / Azimuth Calculation

Definition

Bearing (or azimuth) is the compass direction from one geographic point to another, measured clockwise from North (0°) through East (90°), South (180°), and West (270°).

Bearing/Azimuth calculation is necessary in navigation, survey layout, military mapping, and infrastructure alignment. In GIS, bearing is computed from the difference in latitude and longitude between two points using trigonometric relationships. The result is expressed as an angle in degrees measured clockwise from true north.

Mathematical Formula

$$\theta = \tan^{-1} \left(\frac{\sin(\Delta\lambda) \cos \phi_2}{\cos \phi_1 \sin \phi_2 - \sin \phi_1 \cos \phi_2 \cos(\Delta\lambda)} \right)$$

$$\text{Bearing} = \left(\frac{180}{\pi} \theta + 360 \right) \bmod 360$$

Where ϕ_1 , ϕ_2 are latitudes in radians and $\Delta\lambda = \lambda_2 - \lambda_1$ is the longitude difference in radians.

Excel Implementation

1. using xlookup for latitude and longitudes of target city.
2. Convert latitude, longitude to radians using RADIANS().
3. Compute X_component: =SIN(RADIANS(Lon2-Lon1))*COS(RADIANS(Lat2))
- 4.. Compute Y_component:
=COS(RADIANS(Lat1))*SIN(RADIANS(Lat2))-SIN(RADIANS(Lat1))*COS(RADIANS(Lat2))*COS(RADIANS(Lon2-Lon1))
5. Bearing : =MOD(DEGREES(ATAN2(Y_component, X_component))+360, 360) and use this formula in excel

Excel Formula

```
=MOD(DEGREES(ATAN2(COS(RADIANS(Lat1))*SIN(RADIANS(Lat2))-SIN(RADIANS(Lat1))*COS(RADIANS(Lat2))*COS(RADIANS(Lon2-Lon1)),  
SIN(RADIANS(Lon2-Lon1))*COS(RADIANS(Lat2))))+360, 360)
```

6. Now using IFS formula to get direction of the city with respect to the target city

Excel Formula

```
=IFERROR(  
IFS(F2="Current City", "No Direction",  
OR(F2>=337.5,F2<22.5),"N",  
F2<67.5,"NE",  
F2<112.5,"E",  
F2<157.5,"SE",  
F2<202.5,"S",  
F2<247.5,"SW",  
F2<292.5,"W",  
F2<337.5,"NW"  
), "No Direction"  
)
```

Sample Results

Bearing w.r.t. city	
City	Kanpur
Latitude	26.3704694
Longitude	80.2925157

State	District	Latitude	Longitude	Distance(kM)	Bearing(degree)	Direction
Andaman and Nicobar	Andaman Islands	12.38257107	92.8229105	2033.296405	137.5152419	SE
Andaman and Nicobar	Nicobar Islands	7.83529075	93.5116012	2489.690775	143.5067795	SE
Andhra Pradesh	Adilabad	19.28451395	78.8132123	802.3473805	191.1863101	S
Andhra Pradesh	Anantapur	14.31206556	77.4601584	1372.8003	192.9402667	S
Andhra Pradesh	Chittoor	13.33109263	78.927639	1456.880598	185.8684596	S
Andhra Pradesh	Cuddapah	14.49028913	78.6961387	1331.394709	187.4702904	S
Andhra Pradesh	East Godavari	16.78271832	82.2432075	1084.95843	168.9130905	S
Andhra Pradesh	Guntur	15.88492577	80.5865761	1166.336497	178.4462147	S
Andhra Pradesh	Hyderabad	17.38971556	78.4666889	1016.177118	191.0366822	S
Andhra Pradesh	Karimnagar	18.51548319	79.4376543	877.8317059	185.9120655	S
Andhra Pradesh	Khammam	17.68392133	80.6700864	966.6816093	177.6193834	S
Andhra Pradesh	Krishna	16.14387275	81.1480506	1140.583276	175.3802049	S
Andhra Pradesh	Kurnool	15.43759715	77.9008684	1240.696083	191.997241	S

Fig 4: Excel sheet showing Table with Bearing and direction of districts w.r.t. to target district

4.7 Conditional Formating

Definition

Conditional formatting in Excel functions similarly to **attribute-based querying** in GIS. Instead of spatially selecting features, it uses logical conditions to filter and highlight records that meet specific criteria.

Conditional formatting serves as a practical substitute for GIS attribute queries when working with spatial data in Excel. It allows users to filter and highlight specific records such as Government-owned Clinics in the Ashanti Region by applying logical AND/OR conditions across multiple attribute columns. Excel highlight the result matching **rows in a table** that satisfy all defined criteria. The **AND()** formula acts like a **WHERE clause** in GIS or SQL, checking multiple attribute conditions simultaneously across the dataset. In QGIS, the same filtering can be achieved through: **Select by Expression** tool.

Excel Implementation

1. Imported the health facility dataset into Excel, containing fields like Region, District, Facility Name, Type, Ownership, Latitude, and Longitude.
2. Created a separate condition table (columns L–M) with three filter criteria:
Ownership : Government
Type : Clinic
Region : Ashanti

conditions	
Ownership	Government

Type	Clinic
Region	ashanti

3. Selected the data range `A\$2: H\$2757` and opened Conditional Formatting > New Rule.

4. Used a formula-based rule with AND logic to highlight rows matching all three conditions simultaneously:

Excel Formula

=AND(\$Region_col=\$M\$5, \$type_col=\$M\$4,\$FOwnership_col=\$M\$3)

Where **\$M\$5** = Region, **\$M\$4** = Type, and **\$M\$3** = Ownership reference cells.

5. Applied *Yellow Fill with Dark Yellow Text* to visually distinguish the filtered facilities from the rest of the dataset.

6. Result Interpretation Rows satisfying all three conditions got highlighted.

7. select all the rows(i.e. A:H) then click on **Sort and Filter** then **Custom Sort** then in the **sort on** column drag down to **Cell color** and then in color/icon column select the **light color**.

Sample Results

Region	District	FacilityName	Type	Town	Ownership	Latitude	Longitude
Ashanti	Adansi South	Hwidiem Community Clinic	Clinic	Hwidiem	Government	5.89432	-1.46766
Ashanti	Adansi South	Obonsu Clinic	Clinic	Obonsu	Government	5.95324	-1.35225
Ashanti	Adansi South	Atwereboana Clinic	Clinic	Atwereboana	Government	5.96235	-1.34213
Ashanti	Adansi South	Allen Health Clinic	Clinic	Wamasi	Government	6.08691	-1.69553
Ashanti	Obuasi Municipal	Anyinam Clinic	Clinic	Obuasi-Anyinam	Government	6.19897	-1.68205
Ashanti	Amansie Central	Mile 14 Community Clinic	Clinic	Mile 14	Government	6.22126	-1.82139
Ashanti	Amansie Central	Akrofrom Community Clinic	Clinic	Akrofrom	Government	6.39651	-1.60783
Ashanti	Bosome Freho	Gyasikrom Health Centre	Clinic	Gyasikrom	Government	6.40145	-1.47313
Ashanti	Bosome Freho	Duasi Community Clinic	Clinic	Duasi	Government	6.46863	-1.39522
Ashanti	Asante-Akim South	Banso SDA Clinic	Clinic	Banso	Government	6.47584	-1.19674
Ashanti	Bosome Freho	Detieso Community Clinic	Clinic	Detieso	Government	6.47674	-1.44168
Ashanti	Kumasi Metropolitan	Sepe-Buokrom Community Clinic	Clinic	Buokrom	Government	6.72118	-1.59593
Ashanti	Ejisu-Juaben Municipa	Tikrom Clinic	Clinic	Tikrom	Government	6.73154	-1.51337
Ashanti	Asante-Akim North Mu	Nutrition Rehabilitation Centre	Clinic	Konongo	Government	6.79529	-1.08871
Ashanti	Atwima-Nwabiagya	Atwima Clinic	Clinic	Atwima	Government	7	-2
Ashanti	Sekyer Central	Jeduako Clinic	Clinic	Jeduako	Government	7.09501	-1.20917
Ashanti	Ejura Sekyeredumasi	Babaso MCH	Clinic	Babaso	Government	7.34065	-1.32417
Ashanti	Ejura Sekyeredumasi	Ejura MCH	Clinic	Ejura	Government	7.36556	-1.34358
Ashanti	Ejura Sekyeredumasi	Homako Community Clinic	Clinic	Homako	Government	7.42915	-1.45106
Ashanti	Ejura Sekyeredumasi	Zambrama Community Clinic	Clinic	Zambrama	Government	7.52022	-1.42013
Ashanti	Asante-Akim South	Morso Clinic	Clinic	Morso	Government	6.655180000	-1.10681
Ashanti	Kumasi Metropolitan	KMA Clinic	Clinic	Adum	Government	6.684880000	-1.62353
Ashanti	Ejura Sekyeredumasi	Nkwanta Community Clinic	Clinic	Ejura Nkwanta	Government	7.513780000	-1.34725
Brong Ahafo	Asunafo South	Asarekrom Rural Clinic	Clinic	Asarekrom	Government	6.57574	-2.63918
Brong Ahafo	Asutifi	Sienkyemu Rural Clinic	Clinic	Siekyemu	Government	6.70723	-2.39529
Brong Ahafo	Asunafo North Municip	Ayomso Rural Clinic	Clinic	Ayomso	Government	6.76364	-2.58248
Brong Ahafo	Asunafo North Municip	Ayum Forest Clinic	Clinic	Mim	Government	6.90647	-2.56449
Brong Ahafo	Asutifi	Gambia No1 Rural Clinic	Clinic	Gambia No.1	Government	7.0378	-2.6357
Brong Ahafo	Dormaa Municipal	Kwameasua Presby Clinic	Clinic	Kwameasua	Government	7.16715	-2.7701
Brong Ahafo	Dormaa Municipal	Asikasu Rural Clinic	Clinic	Asikasu	Government	7.1893	-2.89066
Brong Ahafo	Dormaa Municipal	Amasu Rural Clinic	Clinic	Ammasua	Government	7.22802	-2.95807
Brong Ahafo	Tano North	Adrobaa Rural Clinic	Clinic	Adrobaa	Government	7.2628	-2.13179
Brong Ahafo	Dormaa East	Kyeremasu Rural Clinic	Clinic	Kyeremasu	Government	7.26312	-2.76114

figure 5 : Excel sheet showing Result table after applying three Conditions and sorting.

4.8 Pictorial Representation

Definition

Pictorial representation refers to the use of charts and graphs to visually summarize attribute data from a GIS dataset, enabling quick pattern recognition across large records without spatial mapping.

The Health data can be analysed using Pictorial Representation by using bar chart, pie chart etc.. The health facility dataset was used to generate a bar chart and pie chart in Excel to display the distribution of facility types and ownership categories. The bar chart shows the count of each facility type such as Clinic, Health Centre, Hospital, and CHPS while the pie chart shows the proportional share of ownership types including Government, Private, CHAD, and NGO.

Excel Implementation

1. Different type of facility and ownership is observed then by using COUNTIF function, total count of each facility and ownership is calculated.

Excel Formula

```
=COUNTIF(D2:D2857,"Clinic")
```

2. In the **Insert** tab, Bar chart and Pie chart can be selected out of wide range. Then select the data by using which bar chart and pie chart will show the results.

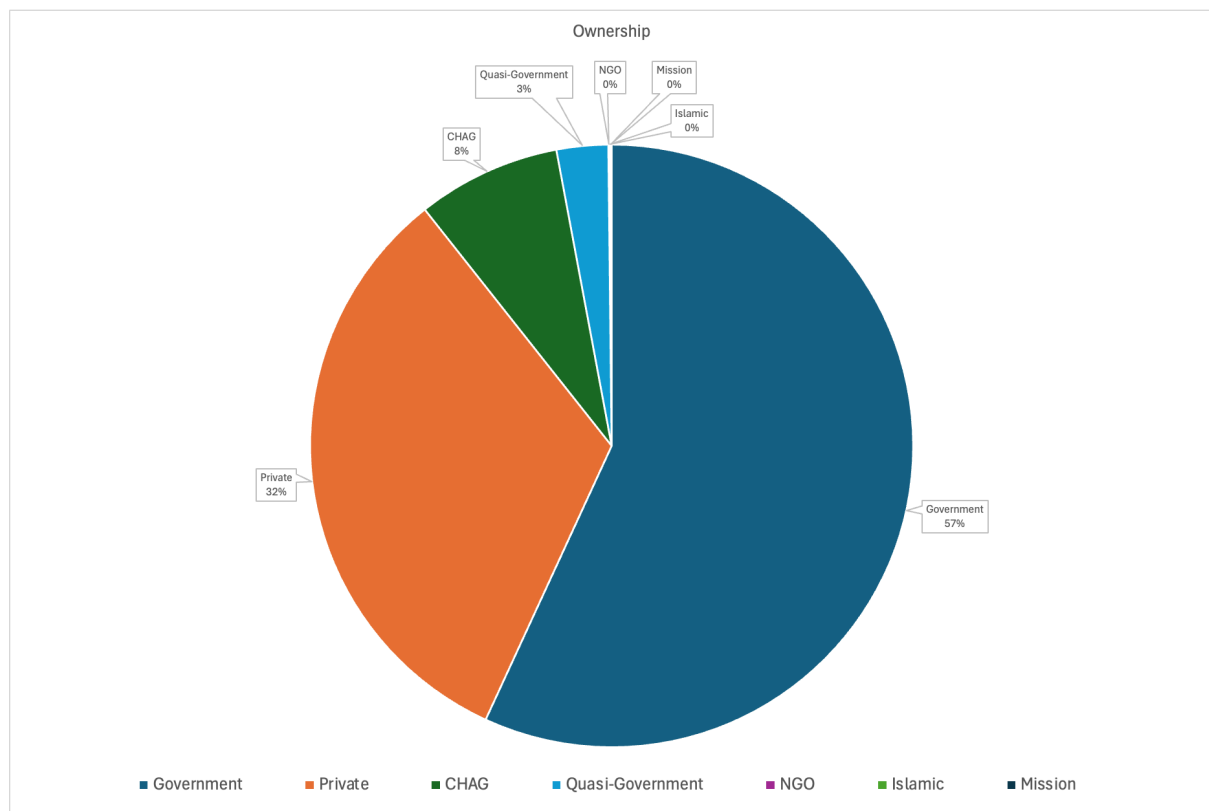
Sample Results

Figure 6 :Pie chart of ownership of health facility

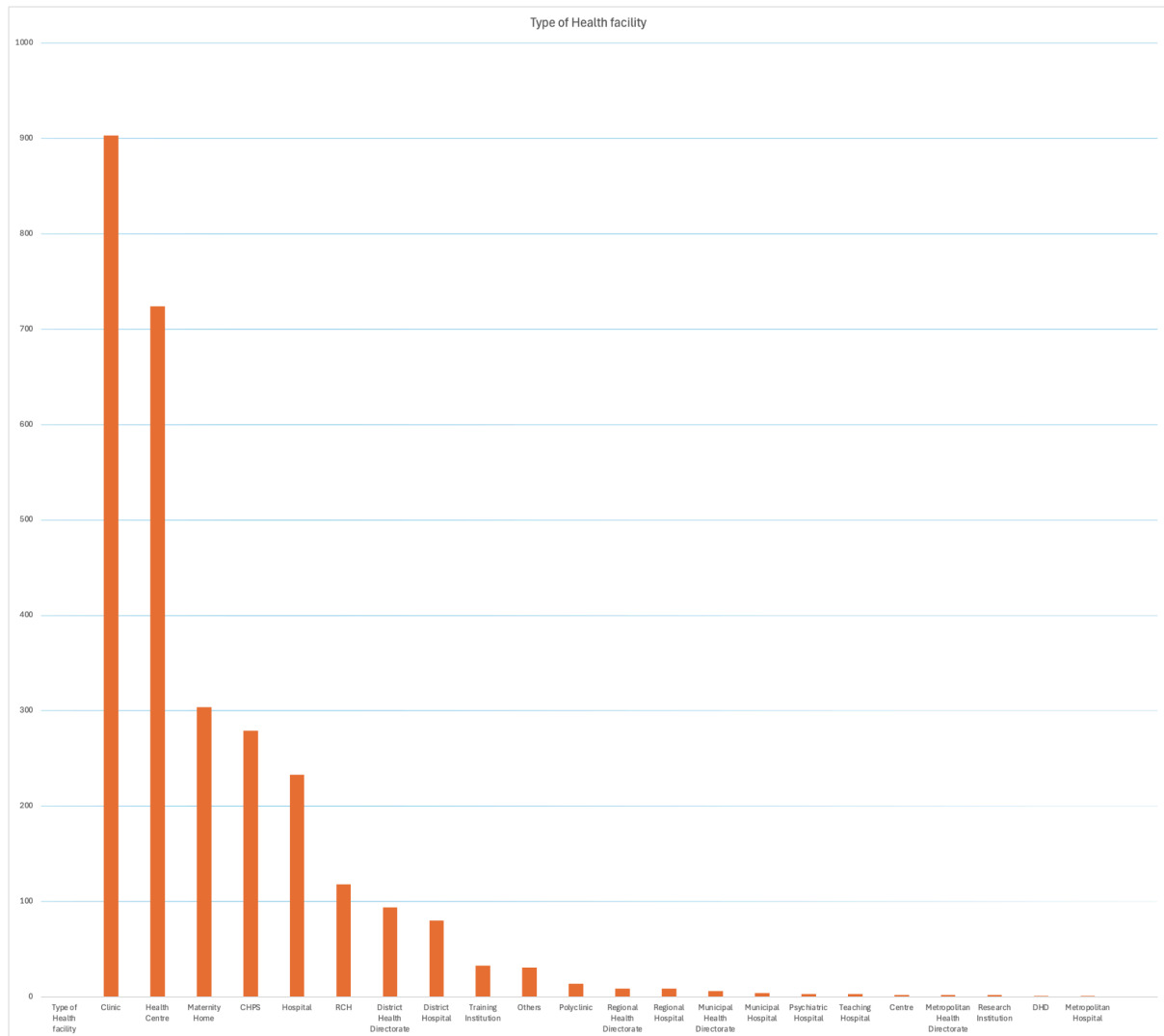


Figure 7 : Bar chart of Type of Health facility

4.9 Choropleth Map using Excel 3D Maps (Bing Maps)

Definition

A choropleth map is a thematic map where regions are shaded in proportion to the value of a statistical variable. In the below map, total traffic accident cases per state are shown. Darker shades indicate higher values and lighter shades indicate lower values.

Excel Implementation

1. Two columns of state-wise traffic accident data namely State name and Total Traffic Accident Cases compiled in Excel. Ensured state names matched standard geographic names recognized by Bing Maps to allow correct

boundary mapping.

2. Used Insert > Maps > Filled Map option in Excel, which automatically added state names to the India base map and plotted them.
3. The accident count values were mapped to a blue gradient color scale. Lighter blue for low values and dark navy for high values making spatial patterns visible.
4. State-wise case numbers were displayed directly on each region for quick reference to avoid separate lookups.

Sample Results

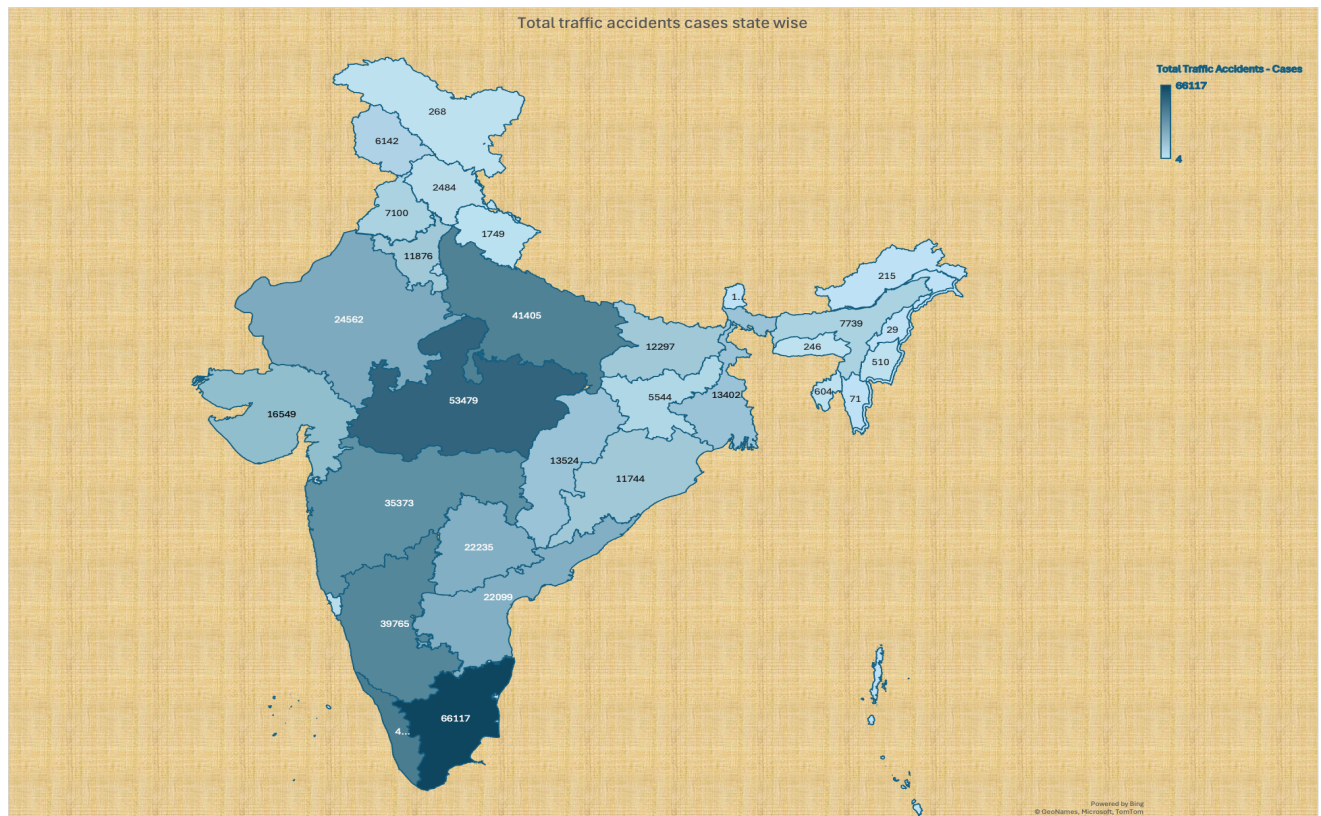


figure 8 :Choropleth Map showing total traffic accidents cases per state

4.10 Coordinate Mapping using ArcGIS for Excel(Add-in)

Definition

Coordinate mapping is the process of plotting geographic point locations on a map using their latitude and longitude values. ArcGIS for Excel is an Esri add-in that enables spatial visualization of tabular coordinate data directly within the Excel environment using interactive Esri basemaps.[12]

Excel Implementation

1. Tabular data containing State, District, Latitude and Longitude columns was arranged in Excel. Each row represented one district with its corresponding centroid coordinates.

2. ArcGIS for Excel was enabled through *Insert > Add-ins > Get Add-ins* from the Excel ribbon and signed in as a Guest.
3. The coordinate columns were selected and chosen into the ArcGIS map panel, specifying which columns represent latitude and longitude fields.
4. The add-in rendered all district points as circular markers over an Esri world basemap, with India's boundary clearly visible and all state clusters identifiable.
5. Marker size, color, and basemap style were adjusted for better visual clarity and presentation and map extent was locked using navigation tools in toolbars.

Sample Results

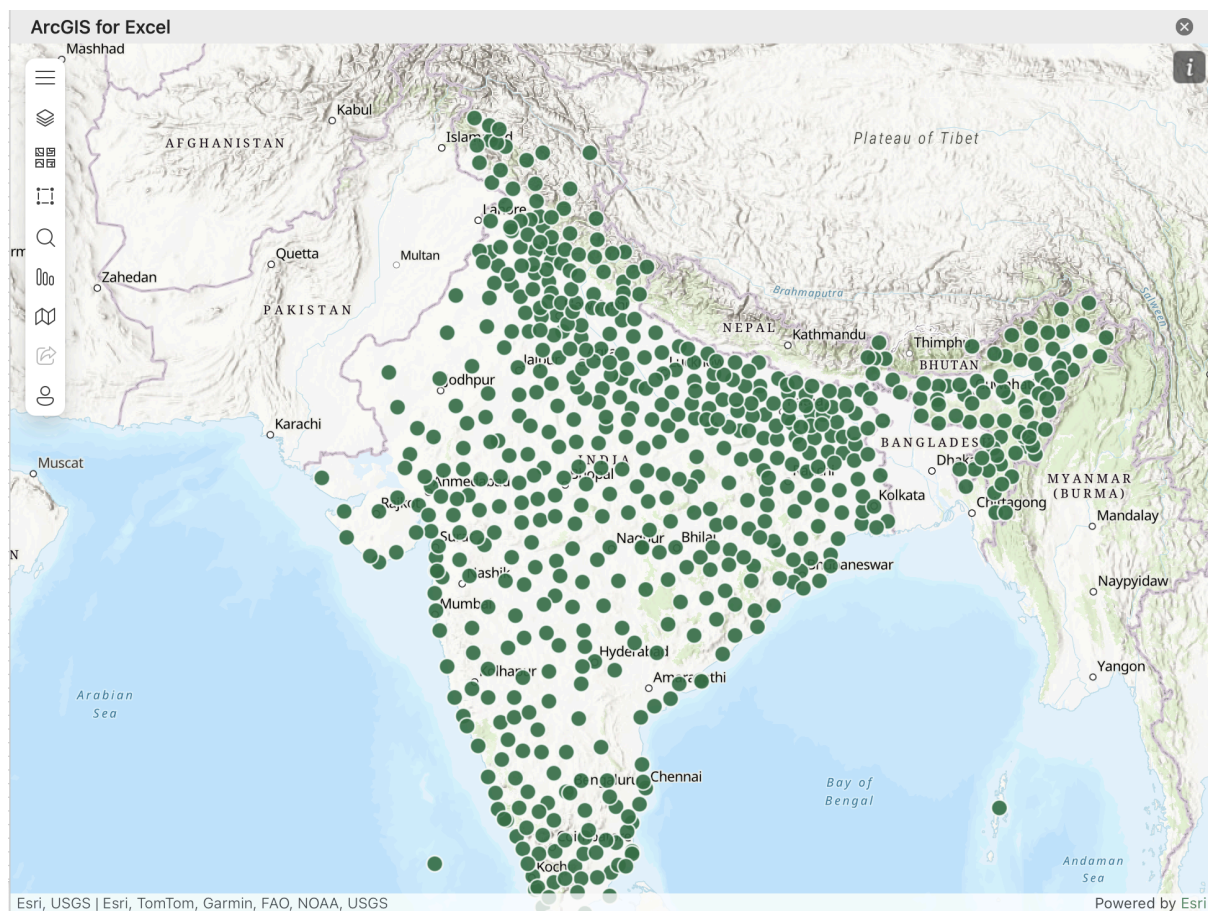


figure 9 :Coordinate mapping of districts of India using ArcGIS for Excel Add-in

4.11 : HeatMap Visualization

Definition

A visualization technique where data values are represented using color variations instead of numbers.

1. Select the range of value or say column data on which we want to apply heatmap.
2. Go to Home > Conditional Formatting > Color Scales. Choose Red - Yellow - Green Color Scale.

Road Accidents by State											Summary		
Sl. No.	State	Accident Details				Accident Details				Total Traffic Accidents - Cases	Total Traffic Accidents - Injured	Total Traffic Accidents - Died	
		Road Accident	Road Accident	Road Accident	Road Accident	Road Accident	Road Accident	Road Accident	Road Accident				
1	Andhra Pradesh	21070	21340	8293	1029	0	1037	0	0	22099	21340	9330	
2	Arunachal Pradesh	215	177	152	0	0	0	0	0	215	177	152	
3	Assam	7028	5679	3060	689	0	693	22	0	7739	5679	3775	
4	Bihar	10801	7068	8898	1444	6	1446	52	0	12297	7074	10397	
5	Chhattisgarh	13091	11459	5890	429	21	429	4	0	13524	11480	6323	
6	Goa	3012	1071	274	54	0	54	0	0	3066	1071	328	
7	Gujarat	15777	15139	7634	766	0	777	6	0	16549	15139	8417	
8	Haryana	10654	8353	5228	1127	26	1101	95	0	11876	8379	6424	
9	Himachal Pradesh	2484	3891	979	0	0	0	0	0	2484	3891	979	
10	Jharkhand	5175	3747	3898	362	19	346	7	0	5544	3766	4251	
11	Karnataka	39765	48154	11705	0	0	0	0	0	39765	48154	11705	
12	Kerala	43592	49317	4317	194	0	194	184	1	43970	49318	4696	
13	Madhya Pradesh	51965	51264	13918	1275	0	1275	239	0	53479	51264	15432	
14	Maharashtra	30546	23878	15748	4827	1686	3145	0	0	35373	25564	18893	
15	Manipur	508	817	127	2	0	2	0	0	510	817	129	
16	Meghalaya	246	310	162	0	0	0	0	0	246	310	162	
17	Mizoram	71	28	83	0	0	0	0	0	71	28	83	
18	Nagaland	24	42	22	0	0	0	5	0	29	42	27	

5. Discussion

- Distance Calculation can be used in various types of analysis such as pipeline route optimization, highway planning, rail or metro corridor analysis etc.
- Nearest neighbor is used to planning of facility which would be required by us in the near future such as (nearest hospital, fire station, police station, railway station, school etc.).
- For road and highway design, drainage channel design (minimum slope for flow), and erosion susceptibility mapping.
- Survey traverse calculations, road alignment bearing determination, pipeline routing, and military/civil defense mapping.
- Pictorial representation helps in planning infrastructure by identifying dominant facility types in a region, which one must keep in consideration regarding the facility.
- Identifying accident-prone highway corridors for road safety intervention.
- By using ArcGIS Add-in, we can do Quick spatial verification of survey coordinate datasets. We can do Preliminary site mapping before detailed GIS analysis.

- For distance calculation here we are assuming a perfect sphere. For high-accuracy engineering surveys, the WGS84 ellipsoid model (used in QGIS/ArcGIS) gives more precise results, especially over long distances.
- We can find the nearest neighbor among a finite set of points in excel, but cannot perform true k-d tree spatial indexing or handle thousands of candidates efficiently.

- In excel we are computing slope by just two discrete points where as a true slope map requires raster DEM processing across thousands of grid cells simultaneously, which surely requires ArcGIS or QGIS Raster Analysis.
- Pictorial representations are non-spatial and they show counts but cannot display geographic distribution.
- Excel maps uses Bing geocoding so sometimes unrecognized or misspelled names won't plot. We cannot perform spatial analysis like hotspot detection or kernel density.
- We need to have an active internet connection to use ArcGIS. There is no support provided for shapefile or raster data integration.
- Excel cannot highlight the matched data on a map and filtered rows have no geographic visualization.

6. Conclusion

Working through eleven GIS functions in Microsoft Excel turned out to be more revealing than we initially expected not just in terms of what Excel *can* do, but perhaps more importantly, in terms of *why* dedicated GIS tools exist in the first place.

The Haversine formula, bearing calculations, slope estimation, and proximity queries all translated surprisingly well into Excel's formula environment. Being forced to write out every trigonometric step manually rather than clicking a button in QGIS or ArcGIS and actually deepened our understanding of what these operations are really computing under the hood.

The exercise also exposed Excel's hard limits fairly quickly. The moment we needed anything beyond point-to-point arithmetic slope maps, kernel density estimation, true polygon overlays, or shapefile integration Excel simply had no answer. The choropleth map via Bing and the ArcGIS add-in were functional, but both depend on external services and internet connectivity, which somewhat undermines the "self-contained spreadsheet" appeal.

What we came away with is a clearer sense of where Excel genuinely fits in a GIS workflow: early-stage data verification, rapid proximity checks, attribute filtering, and most importantly, as a teaching platform where the mathematics of spatial analysis can be made fully transparent. For a course like CE432, that transparency has real pedagogical value students see the equations, not just the outputs. Excel is not a GIS tool. But used thoughtfully, it can be a remarkably effective lens for understanding GIS better.

Contribution

50% anurag

50% ansh

The following references were consulted in the preparation of this term paper :

1. Traffic accidents data. [Data.gov.in](https://data.gov.in)
2. <https://www.kaggle.com/datasets/sirpunch/indian-census-data-with-geospatial-indexin-g>
3. <https://github.com/Vynex/indian-cities-geodata>
4. <https://www.kaggle.com/datasets/alexisbcook/geospatial-learn-course-data>
5. Moser, J. Choropleth Maps: Critical Reflections on the Frequently Used Representation Method in an Educational Context. *KN J. Cartogr. Geogr. Inf.* 75, 191–204 (2025). <https://doi.org/10.1007/s42489-025-00203-2>
6. Kang-Tsung Chang, Introduction to Geographic Information Systems, 9th ed., McGraw-Hill. https://opac.atmaluhur.ac.id/uploaded_files/temporary/DigitalCollection/NmI5NzkxZmE5YTU0YTVkN2Q2N2YxNzQ3N2QxMjM5MDk3YzlmNzdhMA==.pdf
7. P. A. Longley, M. F. Goodchild, D. J. Maguire, and D. W. Rhind, Geographic Information Science & Systems, 4th ed. Hoboken, NJ: Wiley, 2015.
8. R. W. Sinnott, "Virtues of the Haversine," *Sky and Telescope*, vol. 68, no. 2, p. 159, 1984.
9. ESRI, "An Overview of Spatial Analysis," ArcGIS Pro Documentation, Esri Inc., Redlands, CA, 2023. [Online]. Available: <https://pro.arcgis.com/en/pro-app>
10. J. Goldstein, "Teaching GIS Fundamentals with Spreadsheets," *Journal of Geography in Higher Education*, vol. 44, no. 3, pp. 412–428, 2020.
11. QGIS Development Team, QGIS Geographic Information System, Open Source Geospatial Foundation, 2023. [Online]. Available: <https://qgis.org>
12. Cressie, N. (1993). *Statistics for Spatial Data*. Wiley.
13. Microsoft Corporation, "Excel Mathematical and Trigonometric Functions," Microsoft Office Documentation, 2024. [Online]. Available: <https://support.microsoft.com/en-us/office>
14. Vincenty, T. (1975). "Direct and Inverse Solutions of Geodesics on the Ellipsoid." *Survey Review*, 23(176), 88–93.